

Future Network Architecture Ideas

Network Architecture

- VRF (multi-tenant isolation)
- Shared services network (DNS/AAA accessible from multiple VRFs)
- VXLAN overlays (optional later)
- Service zones (DMZ / App / DB / Mgmt segmentation)
- Spine-leaf architecture (future scaling idea)

Routing & Edge

- OSPF for internal routing
- eBGP to simulated ISPs
- IP SLA + tracked default routes
- BGP failover scenarios
- Route filtering / prefix lists
- ECMP

High Availability

- HSRP tracking
- Load balancer health checks
- Server clustering
- Database replication
- Automatic DB failover
- Reverse proxy layer

Security

- IDS (Suricata / Snort)
- SIEM (ELK stack)
- Web Application Firewall (WAF)
- Rate limiting for authentication endpoints
- Zero-trust segmentation
- DNS security (DNS poisoning detection)
- Network access control (802.1X)

Monitoring & Logging

- Syslog aggregation
- SNMP monitoring
- NetFlow / traffic analysis
- Alert dashboards
- Log integrity checks
- SLA monitoring

Automation & SDN

- Ansible playbooks
- Terraform infrastructure deployment
- Network configuration templates
- Git-based configuration versioning
- NetBox as source of truth
- API-based device configuration
- CI/CD pipeline for network changes

Infrastructure / Cloud

- Hypervisor cluster (Proxmox / KVM)
- IaaS tenant networks
- Self-service provisioning concept
- Storage replication
- Object storage service
- Container workloads (Docker / Kubernetes)

VPN Service Improvements

- Site-to-site VPN
- Remote access VPN
- MFA for authentication
- Per-user logging
- Bandwidth limits
- Abuse detection

Attack / Security Scenarios

- DNS cache poisoning
- ARP spoofing
- Man-in-the-middle attack
- Log tampering
- Privilege escalation
- Ransomware lateral movement
- Data exfiltration simulation

Operational / Reliability Testing

- Link failure tests
- Router failure tests
- Switch failure tests
- ISP outage simulation
- Configuration rollback tests
- Disaster recovery testing